Numeric Integrity Audit

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1 Scope

This report summarizes numeric-integrity checks aligned with Heathers' data-technique family:

- internal rounding consistency for count-percent table cells

- `scrutiny` GRIM / GRIMMER / DEBIT and duplicate-value checks
- rounding-bias diagnostics for reported decimal patterns
- statistical-consistency checks via `statcheck`

2 Load Outputs

```
repo_root <- normalizePath("..", winslash = "/", mustWork = TRUE)
study_id <- Sys.getenv("FORENSICS_STUDY_ID", "lungtime")
report_dir <- file.path(repo_root, "reports", "numeric", study_id)

required <- c(
  "numeric_row_results.csv",
  "numeric_summary.csv",
  "numeric_package_status.csv",
  "numeric_scrutiny_grim_raw.csv",
  "numeric_scrutiny_grim_audit.csv",
  "numeric_scrutiny_grimmer_raw.csv",
  "numeric_scrutiny_grimmer_audit.csv",
  "numeric_scrutiny_debit_raw.csv",
  "numeric_scrutiny_debit_audit.csv",
  "numeric_scrutiny_duplicates.csv",
  "numeric_scrutiny_rounding_bias.csv",
  "numeric_statcheck_raw.csv",
  "numeric_standardized_results.csv"
)
missing <- required[!file.exists(file.path(report_dir, required))]
if (length(missing) > 0) {
  stop("Missing files: ", paste(missing, collapse = ", "))
}

row_results <- read.csv(file.path(report_dir, "numeric_row_results.csv"), check.names = FALSE
summary_tbl <- read.csv(file.path(report_dir, "numeric_summary.csv"), check.names = FALSE)
pkg_status <- read.csv(file.path(report_dir, "numeric_package_status.csv"), check.names = FA
grim_raw <- read.csv(file.path(report_dir, "numeric_scrutiny_grim_raw.csv"), check.names = F
grim_audit <- read.csv(file.path(report_dir, "numeric_scrutiny_grim_audit.csv"), check.names
grimmer_raw <- read.csv(file.path(report_dir, "numeric_scrutiny_grimmer_raw.csv"), check.name
grimmer_audit <- read.csv(file.path(report_dir, "numeric_scrutiny_grimmer_audit.csv"), check
debit_raw <- read.csv(file.path(report_dir, "numeric_scrutiny_debit_raw.csv"), check.names =
debit_audit <- read.csv(file.path(report_dir, "numeric_scrutiny_debit_audit.csv"), check.name
duplicates_raw <- read.csv(file.path(report_dir, "numeric_scrutiny_duplicates.csv"), check.na
```

```r
rounding_bias <- read.csv(file.path(report_dir, "numeric_scrutiny_rounding_bias.csv"), check
statcheck_raw <- read.csv(file.path(report_dir, "numeric_statcheck_raw.csv"), check.names = F
standardized <- read.csv(file.path(report_dir, "numeric_standardized_results.csv"), check.nam

compact_character <- function(x, width = 48) {
  x <- as.character(x)
  x[is.na(x)] <- ""
  too_long <- nchar(x) > width
  x[too_long] <- paste0(substr(x[too_long], 1, width - 3), "...")
  x
}

compact_df <- function(df, cols = NULL, n = 20, char_width = 48) {
  out <- df
  if (!is.null(cols)) {
    keep <- intersect(cols, names(out))
    out <- out[, keep, drop = FALSE]
  }
  if (nrow(out) > n) {
    out <- utils::head(out, n)
  }
  is_char_like <- vapply(out, function(col) is.character(col) || is.factor(col), logical(1))
  out[is_char_like] <- lapply(out[is_char_like], compact_character, width = char_width)
  out
}

render_table <- function(df, note, cols = NULL, n = 20, digits = 4, char_width = 48) {
  if (nrow(df) > 0) {
    knitr::kable(compact_df(df, cols = cols, n = n, char_width = char_width), digits = digits
  } else {
    cat(note)
  }
}
```

3 Section 4 Summary Metrics Key

These are screening indicators from Heathers' data-techniques framework, not proof of misconduct.

- `n_rows`: number of categorical count-percent cells checked for arithmetic consistency.
- `n_rounding_flags`: rows with `|reported_percent - computed_percent| >= 0.2`.

- `rounding_flag_rate`: `n_rounding_flags / n_rows`; larger values suggest widespread reporting drift.
- `median_abs_percent_delta`, `max_abs_percent_delta`: typical and worst discrepancy magnitude.
- `scrutiny_cases_n`: total canonical rows prepared for scrutiny-family methods.
- `grim_cases`, `grim_incons_cases`: testable rows and inconsistency count from `scrutiny::grim_map`.
- `grimmer_cases`, `grimmer_incons_cases`: testable rows and inconsistency count from `scrutiny::grimmer_map`.
- `debit_cases`, `debit_incons_cases`: binary-proportion rows testable by DEBIT and their inconsistency count.
- `duplicate_flag_rows`: rows with at least one duplicated field (`x`, `sd`, or `n`) in duplicate checks.
- `rounding_bias_groups`: `(trial_id, digits_x)` groups assessed for directional rounding imbalance.
- `statcheck_cases`, `statcheck_errors`, `statcheck_decision_errors`: parsed test rows, any inconsistencies, and significance-changing inconsistencies.
- `scrutiny_seq_enabled`: whether optional sequence-space checks were executed.

Empty method tables can mean "no eligible rows", "package unavailable", or execution failure; inspect `numeric_package_status.csv` execution notes for attribution.

```r
if (nrow(summary_tbl) > 0) {
  summary_long <- data.frame(
    metric = names(summary_tbl),
    value = as.character(summary_tbl[1, ]),
    stringsAsFactors = FALSE
  )
  knitr::kable(summary_long)
} else {
  cat("No summary rows available.")
}
```

metric	value
trial_id	pronto_lrm_2026_s2213260025004333
n_rows	86
n_rounding_flags	0
rounding_flag_rate	0
n_reported_p	0
median_abs_percent_delta	0.0275748721694674
max_abs_percent_delta	0.0499634769905039
scrutiny_cases_n	0

metric	value
grim_available	TRUE
grim_cases	0
grim_incons_cases	0
grimmer_available	TRUE
grimmer_cases	0
grimmer_incons_cases	0
debit_available	TRUE
debit_cases	0
debit_incons_cases	0
duplicates_available	TRUE
duplicate_flag_rows	0
rounding_bias_available	TRUE
rounding_bias_groups	0
statcheck_available	TRUE
statcheck_rows	0
statcheck_cases	0
statcheck_errors	0
statcheck_decision_errors	0
rsprite2_rows	43
scrutiny_seq_enabled	FALSE

4 Package Availability

```r
render_table(
  pkg_status,
  "No package-status output.",
  cols = c("package", "installed", "version", "objective", "executed", "execution_note"),
  n = 20,
  char_width = 42
)
```

package	installed	version	objective	executed	execution_note
scrutiny	TRUE	0.6.1	GRIM/GRIMMER/DEBIT/duplication/rounding…	TRUE	grim= No GRIM-eligible rows. ; grimmer=…

package	installed	version	objective	executed	execution_note
rsprite2	FALSE		SPRITE-style proportion and distributio…	FALSE	Package execution not yet implemented i…
statcheck	TRUE	1.5.0	Recompute p-values from reported statis…	TRUE	ok

5 GRIM Results

```
render_table(
  grim_audit,
  "No GRIM audit output (package unavailable or no GRIM-eligible rows).",
  n = 20
)
```

```
No GRIM audit output (package unavailable or no GRIM-eligible rows).
```

```
render_table(
  grim_raw,
  "No GRIM row-level output.",
  cols = c("case_id", "source_unit", "x", "n", "consistency", "probability"),
  n = 20
)
```

```
No GRIM row-level output.
```

6 GRIMMER Results

```
render_table(
  grimmer_audit,
  "No GRIMMER audit output (package unavailable or no GRIMMER-eligible rows).",
  n = 20
)
```

```
No GRIMMER audit output (package unavailable or no GRIMMER-eligible rows).
```

```
render_table(
  grimmer_raw,
  "No GRIMMER row-level output.",
  cols = c("case_id", "variable", "level", "group", "x", "sd", "n", "consistency", "reason")
  n = 20,
  char_width = 36
)
```

```
No GRIMMER row-level output.
```

7 DEBIT Results

```
render_table(
  debit_audit,
  "No DEBIT audit output (package unavailable or no DEBIT-eligible rows).",
  n = 20
)
```

```
No DEBIT audit output (package unavailable or no DEBIT-eligible rows).
```

```
render_table(
  debit_raw,
  "No DEBIT row-level output.",
  cols = c("case_id", "source_unit", "x", "sd", "n", "consistency", "rounding"),
  n = 20,
  char_width = 40
)
```

```
No DEBIT row-level output.
```

8 Duplication And Rounding-Bias Results

```
render_table(
  duplicates_raw,
  "No duplicate-check rows available.",
```

```
  cols = c("case_id", "variable", "level", "group", "x_dup", "sd_dup", "n_dup", "x_n", "sd_n
  n = 20,
  char_width = 36
)
```

```
No duplicate-check rows available.
```

```
render_table(
  rounding_bias,
  "No rounding-bias groups available.",
  cols = c("trial_id", "digits_x", "n_values", "bias_up", "bias_down", "abs_bias_gap", "anoma
  n = 20
)
```

```
No rounding-bias groups available.
```

9 Statcheck Results

```
if (nrow(statcheck_raw) > 0) {
  statcheck_ranked <- statcheck_raw[order(statcheck_raw$error, decreasing = TRUE), ]
  knitr::kable(
    compact_df(
      statcheck_ranked,
      cols = c(
        "source", "test_type", "df1", "df2", "test_value",
        "reported_p", "computed_p", "error", "decision_error"
      ),
      n = 20,
      char_width = 36
    ),
    digits = 4
  )
} else {
  cat("No statcheck output (package unavailable or no parseable report text).")
}
```

```
No statcheck output (package unavailable or no parseable report text).
```

10 Standardized Output Snapshot

```r
render_table(
  standardized,
  "No standardized rows available.",
  cols = c(
    "trial_id", "method", "metric", "value_numeric",
    "p_value", "anomaly_flag", "severity", "details"
  ),
  n = 20,
  digits = 4,
  char_width = 40
)
```

trial_id	method	metric	value_numeric	p_value	anomaly_flag	severity	details
pronto_lrm_2026_s02213260025004333	reporting_consistency	abs_percent_delta	0.0142	NA	FALSE	low	reported_percent=84.5; computed_perce...
pronto_lrm_2026_s02213260025004333	reporting_consistency	abs_percent_delta	0.0302	NA	FALSE	low	reported_percent=85.2; computed_perce...
pronto_lrm_2026_s02213260025004333	reporting_consistency	abs_percent_delta	0.0310	NA	FALSE	low	reported_percent=82.5; computed_perce...
pronto_lrm_2026_s02213260025004333	reporting_consistency	abs_percent_delta	0.0466	NA	FALSE	low	reported_percent=82.9; computed_perce...
pronto_lrm_2026_s02213260025004333	reporting_consistency	abs_percent_delta	0.0348	NA	FALSE	low	reported_percent=0.4; computed_percen...
pronto_lrm_2026_s02213260025004333	reporting_consistency	abs_percent_delta	0.0475	NA	FALSE	low	reported_percent=0.6; computed_percen...
pronto_lrm_2026_s02213260025004333	reporting_consistency	abs_percent_delta	0.0000	NA	FALSE	low	reported_percent=0; computed_percent=0

trial_id	method	metric	value_numeric	p_value	anomaly_flag	severity	details
pronto_lrm_2026_s2213260025004333	reporting_consistency	abs_percent_delta	0.0368	NA	FALSE	low	reported_percent=0; computed_percent=...
pronto_lrm_2026_s2213260025004333	reporting_consistency	abs_percent_delta	0.0199	NA	FALSE	low	reported_percent=1.7; computed_percen...
pronto_lrm_2026_s2213260025004333	reporting_consistency	abs_percent_delta	0.0057	NA	FALSE	low	reported_percent=1.7; computed_percen...
pronto_lrm_2026_s2213260025004333	reporting_consistency	abs_percent_delta	0.0209	NA	FALSE	low	reported_percent=0.6; computed_percen...
pronto_lrm_2026_s2213260025004333	reporting_consistency	abs_percent_delta	0.0262	NA	FALSE	low	reported_percent=0.6; computed_percen...
pronto_lrm_2026_s2213260025004333	reporting_consistency	abs_percent_delta	0.0096	NA	FALSE	low	reported_percent=0.1; computed_percen...
pronto_lrm_2026_s2213260025004333	reporting_consistency	abs_percent_delta	0.0158	NA	FALSE	low	reported_percent=0.2; computed_percen...
pronto_lrm_2026_s2213260025004333	reporting_consistency	abs_percent_delta	0.0461	NA	FALSE	low	reported_percent=0.1; computed_percen...
pronto_lrm_2026_s2213260025004333	reporting_consistency	abs_percent_delta	0.0105	NA	FALSE	low	reported_percent=0.1; computed_percen...
pronto_lrm_2026_s2213260025004333	reporting_consistency	abs_percent_delta	0.0096	NA	FALSE	low	reported_percent=0.1; computed_percen...
pronto_lrm_2026_s2213260025004333	reporting_consistency	abs_percent_delta	0.0263	NA	FALSE	low	reported_percent=0.1; computed_percen...
pronto_lrm_2026_s2213260025004333	reporting_consistency	abs_percent_delta	0.0443	NA	FALSE	low	reported_percent=0.3; computed_percen...
pronto_lrm_2026_s2213260025004333	reporting_consistency	abs_percent_delta	0.0422	NA	FALSE	low	reported_percent=0.3; computed_percen...

11 Largest Rounding Deltas

```
ranked <- row_results[order(-row_results$abs_percent_delta, na.last = TRUE), ]
knitr::kable(head(ranked[, c(
  "variable", "level", "group", "reported_percent", "computed_percent", "abs_percent_delta"
)], 15), digits = 4)
```

	variable	level	group	reported_percent	computed_percent	abs_percent_delta
63	Comorbidities	Moderate to severe CKD (Grade III or above)	pct	13.6	13.5500	0.0500
60	Comorbidities	COPD	usual_care	22.7	22.6519	0.0481
31	Ethnicity	Any other Asian background	pct	0.5	0.5478	0.0478
6	Ethnicity	White Irish	usual_care	0.6	0.5525	0.0475
28	Ethnicity	Bangladeshi	usual_care	0.1	0.1473	0.0473
30	Ethnicity	Chinese	usual_care	0.1	0.1473	0.0473
34	Ethnicity	Black	usual_care	0.8	0.8471	0.0471
4	Ethnicity	White English/Welsh/Scottish/Northern Irish/British	usual_care	82.9	82.9466	0.0466
57	Comorbidities	Diabetes Mellitus	pct	22.8	22.7538	0.0462
15	Ethnicity	White and Black African	pct	0.1	0.1461	0.0461
27	Ethnicity	Bangladeshi	pct	0.1	0.1461	0.0461
37	Ethnicity	Caribbean	pct	0.1	0.1461	0.0461
68	Comorbidities	Stroke or TIA	usual_care	10.8	10.7551	0.0449
61	Comorbidities	Heart Failure	pct	12.3	12.3448	0.0448
53	Number of Listed Comorbidities	2+	pct	58.1	58.1446	0.0446

12 Diagnostics

```
hist(
  row_results$abs_percent_delta,
  breaks = 10,
  col = "gray80",
```

```
  border = "white",
  main = "Absolute Percent Delta Distribution",
  xlab = "|reported_percent - computed_percent|"
)
abline(v = 0.2, col = "red", lwd = 2, lty = 2)
```